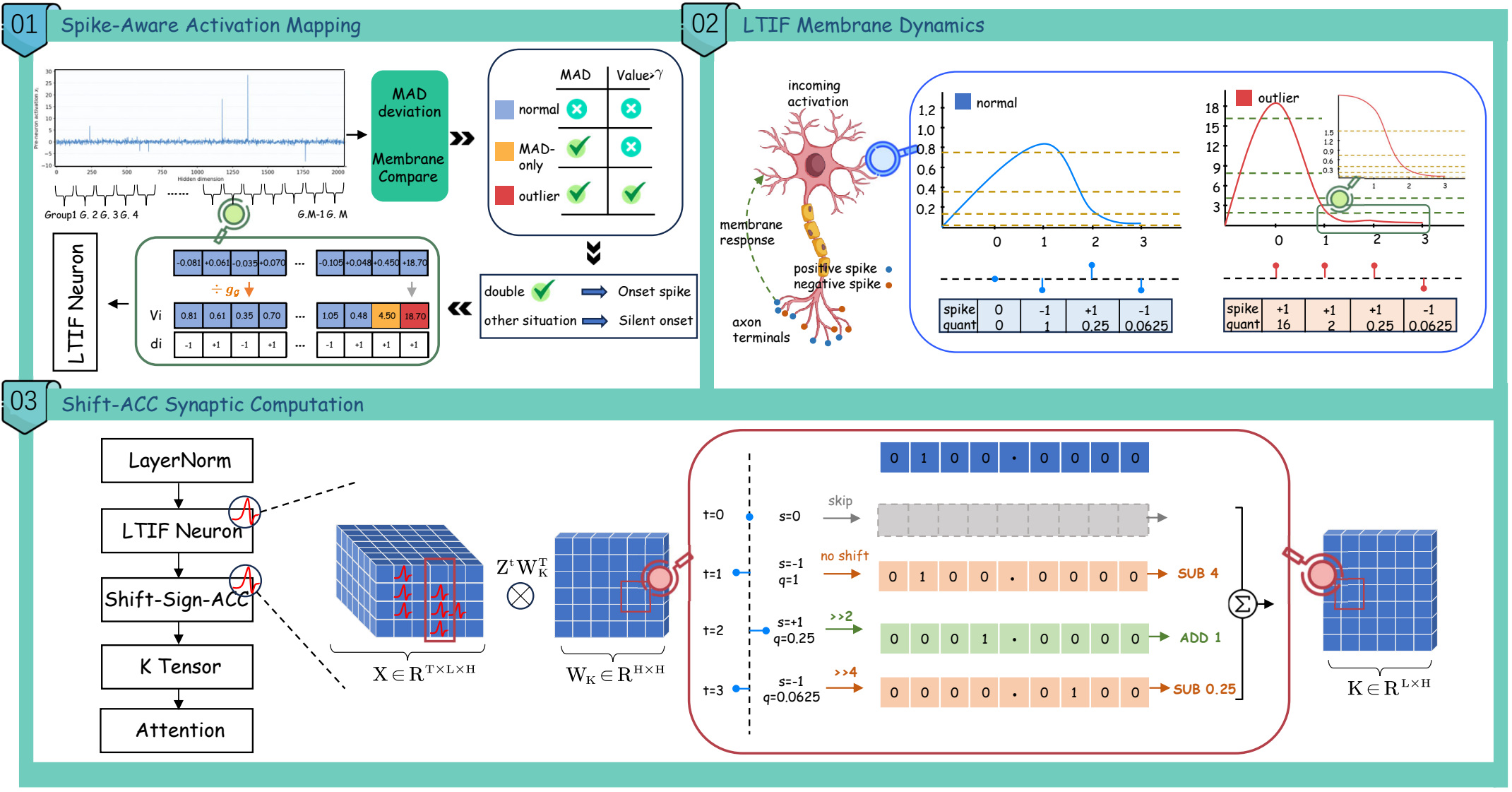

01
Spike-Aware Activation Mapping
MAD deviation
Membrane Compare
MAD
Value>γ
normal
MAD-only
outlier
Group1 G. 2 G. 3 G. 4
G.M-1 G. M
LTIF Neuron
Vi
di
double
Onset spike
other situation
Silent onset
02
LTIF Membrane Dynamics
incoming activation
membrane response
positive spike
negative spike
axon terminals
normal
outlier
spike quant
03
Shift-ACC Synaptic Computation
LayerNorm
LTIF Neuron
Shift-Sign-ACC
K Tensor
Attention
t=0
t=1
t=2
t=3
skip
no shift
SUB 4
ADD 1
SUB 0.25